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Dissertation Brief

Machine Learning-Based Battery Health Prediction and Preventive Maintenance Framework for Electric Vehicle Fleets

Overview

This dissertation aims to develop a data-driven battery diagnostic and preventive maintenance decision framework for EV fleets. As the UK accelerates its transition to electric vehicles, being able to monitor and predict battery health at fleet scale has become a practical and pressing challenge. This research uses Machine Learning to address that — applying classification and prediction models to real battery cycling data, and building a maintenance decision framework from the results.

The study is fully simulation and data-based, using open-source datasets and implemented in Python and MATLAB. No physical lab work or human participants are involved.

Objectives

1. Review existing literature on battery degradation diagnostics, battery management systems, and fleet maintenance strategies in EV applications.
2. Perform data exploration and feature engineering on three open-source battery datasets.
3. Train and comparatively evaluate a minimum of six ML models — Random Forest, XGBoost, Gradient Boosting, SVM, Logistic Regression, and LSTM — using accuracy, F1-score, RMSE, and MAE as evaluation metrics, then justify the selection of the final model(s) based on performance evidence.
4. Develop a capacity degradation prediction model using the best-performing approach from the comparative evaluation (expected: LSTM for time-series sequences) to forecast remaining useful life over charge cycles.
5. Construct a fleet-level preventive maintenance decision framework using the diagnostic outputs from the models.
6. Relate findings to UK EV policy context, including the Zero Emission Vehicle mandate and Faraday Battery Challenge.

Datasets

Three publicly available, peer-validated datasets will be used:

- NASA Prognostics Battery Dataset — lithium-ion charge and discharge cycling data collected under controlled laboratory conditions.

- CALCE Battery Research Group Dataset — capacity fade and impedance measurements across different cell chemistries and cycling conditions.
- Oxford Battery Degradation Dataset — structured ageing experiments with high-fidelity degradation profiles.

Methodology

The work will follow a structured pipeline:

7. Data cleaning and preprocessing — handling noise, normalisation, and cycle segmentation across all three datasets.
8. Feature extraction — deriving State of Health (SoH), internal resistance, capacity fade rate, coulombic efficiency, and cycle count as model inputs.
9. Model comparison — training and tuning six ML models (Random Forest, XGBoost, Gradient Boosting, SVM, Logistic Regression, and LSTM) on the prepared datasets, with hyperparameter optimisation via Grid Search and cross-validation.
10. Model selection and justification — evaluating all models against accuracy, F1-score, RMSE, and MAE; selecting the best-performing model(s) for fault classification and degradation prediction with documented rationale.
11. Framework development — combining model outputs into a maintenance scheduling decision logic applicable at fleet level.
12. Evaluation — using accuracy, F1-score, RMSE, and MAE, with cross-validation to ensure reliability.

Note on Scope and Powertrain Context

While the revised title focuses on ML-based battery health prediction, the real-world context of EV fleet operation remains important to this work: motor current draw from the powertrain directly influences battery stress, ageing behaviour, and fault patterns. Acknowledging this makes the diagnostic framework more representative of how battery health is experienced in a vehicle, rather than in isolation.

However, I am aware that live fleet-level motor current data from operational vehicles is commercially sensitive and is unlikely to be accessible through university channels. I have discussed this with a colleague and want to be transparent about how I plan to address it.

The powertrain-integrated aspect of this dissertation will be treated as a conceptual and framework-level contribution rather than a direct data input. The empirical modelling will use the open-source battery datasets listed above. The influence of motor current on battery behaviour will be contextualised using WLTP drive cycle load profiles and electrochemical current-response characteristics present in the datasets. The dissertation will clearly acknowledge this boundary and identify direct powertrain telemetry integration as a direction for future work.

I would welcome your guidance on whether this framing is appropriate, and whether any adjustment to the scope would be advisable.

Completion Plan

The dissertation will be completed across six phases between May and early September 2026:

Phase	Timeline	Key Activities
Setup & Literature Review	May 2026	Supervisor confirmation, EG7030 registration, literature review, dataset download, Python/MATLAB environment setup.
Data Preparation	Early June 2026	Exploratory data analysis, feature engineering, data cleaning and normalisation across all three datasets.
Model Development	Mid June – July 2026	Comparative evaluation of six ML models (RF, XGBoost, Gradient Boosting, SVM, Logistic Regression, LSTM). Hyperparameter tuning, cross-validation, and evidence-based model selection.
Framework Design	Late July 2026	Preventive maintenance decision framework built from model outputs. Powertrain integration documented at conceptual level.
Results & Analysis	Early August 2026	Results interpreted, visualisations prepared, discussion and limitations drafted.
Writing & Submission	August – Early Sept 2026	Full dissertation written (~12,500 words), supervisor review completed, submitted via UEL portal.